「2018. Fall.

1. Let $A \in M_{n\times n}(\mathbb{R})$ such that $A^2=0$. Prove
$$2\cdot \operatorname{rank} A \le n.$$

Proof. Given $x\in\mathbb{R}^n$, $A(Ax)=A^2x=0$, so $\operatorname{Im}A \subseteq \ker A$. So, $\dim \operatorname{Im} A \le \dim \ker A$.
By the dimension theorem,
$$n = \operatorname{nullity} A + \operatorname{rank} A \ge 2\operatorname{rank} A$$
$\blacksquare$

2. Determine over $\mathbb{R}$,
$$A=\begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$
is diagonalizable.

Proof. The characterstic polynomial of A is
$$\det\begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3)+2\cdot(t-2) = (t-2)(t^2-3t+2) = (t-1)(t-2)(t-2)$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to charateristic polynomial, so we need to check that $(A-I)(A-2I)=0$,
$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$
So, the minimal polynomial of A is $(t-1)(t-2)$, it implies A is diagonalizable, because the minimal polynomial splits over $\mathbb{R}$ and separable. Explicity,
$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \begin{cases} x=-2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \operatorname{span}\left\{\begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}\right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \iff \left\{ x+z=0 \right. \Rightarrow \ker(A-2I) = \operatorname{span}\left\{\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}\right\},$$
Therefore, $A = UDU^{-1}$ where $U=\begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$ and $D=\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

[2018. Fall,

3. Let R be a Commutative Ring and let $x \in R$ such that $x^{2018} = 0$.
prove that $1+x$ is unit.

Proof. observe that

$$(1+x)\cdot(-x^{2017}+x^{2016}-x^{2015}+\cdots-x+1)$$

$$= 1-x+x^2+\cdots+x^{2016}-x^{2017}$$
$$+x-x^2+\cdots-x^{2016}+x^{2017}-x^{2018}$$

$= 1$, Since $x^{2018}=0$. So, $1-x+\cdots+x^{2016}-x^{2017}$ is inverse of $1+x$.

4. Prove that A_4 does not have a subgroup of order 6.

Proof. Since $|S_4 : A_4| = 2$, $|A_4| = 4\cdot 3 = 12$.
Suppose that A_4 has a subgroup H of order 6.
Then, $|A_4 : H| = 12/6 = 2$, So $gH = Hg$ for any $g \in A_4$, it means $H \trianglelefteq A_4$ is normal.
Consider the afforded Homomorphism of A_4 acts A_4/H as left multiplication;

$$\varphi : A_4 \to S_{A_4/H} ;\ g \mapsto (xH \mapsto gxH)$$

And, Since every 3-Cycle $\sigma \in A_4$ Satisfies

$$(\varphi(\sigma))^3 = \varphi(\sigma^3) = \varphi(id) = id$$

and $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$ So $\varphi(\sigma) = id$. It implies all 3-Cycles are contained in $\ker \varphi$. Meanwhile, observe that $(1\ 2\ 3), (1\ 2\ 4) \in A_4$ Satisfies

$$(1\ 2\ 3)^{-1}(1\ 2\ 4)(1\ 2\ 3) = (1\ 3\ 2)(1\ 2\ 4)(1\ 2\ 3) = (1\ 4\ 3)$$

$(1\ 2\ 3)(1\ 4\ 3) = (1\ 4)(2\ 3)$, $\cdots$ in this way, A_4 is generated by all 3-Cycles in A_4, so $\ker \varphi = A_4$. By the first isomorphism theorem $A_4/\ker\varphi \cong \operatorname{Im}\varphi$ is trivial, so the action is trivial,
But, the left coset action is transitive, it cannot be trivial, Contradiction. $\square$

「Similar Problems.

1. Show that A_5 has no subgroup of order 15.

Proof. $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$.

Suppose that there is a subgroup $H < A_5$ of order 15.

By the Sylow Theorem, $\# Syl_3(H) \in \{1\}$ and $\# Syl_5(H) \in \{1\}$, that is,

there are order 3 subgroup $P < H$ and order 5 subgroup $Q < H$,

and both are normal subgroup of H. Since $P \cong C_3$ and $Q \cong C_5$ both are cyclic, let $x \in P$ and $y \in Q$ such that $P = \langle x \rangle$ and $Q = \langle y \rangle$.

Consider the afforded Homomorphism by H acts P by conjugation

$$\varphi : H \to S_P ; h \mapsto (k \mapsto hkh^{-1}).$$

Then, $\operatorname{Ker} \varphi = \{ h \in H ; hkh^{-1} = k \text{ for all } k \in P \} = C_H(P)$ and

$H / \operatorname{Ker} \varphi \cong \operatorname{Im} \varphi \leq \operatorname{Aut}(P)$. But, since $\operatorname{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$,

and $|H|$ has no factor 2, odd number, so $|\operatorname{Ker} \varphi| = |H|$, that is,

$C_H(P) = H$. Now, x and y commute, therefore $|xy| = 15$.

But, A_5 cannot have order 15 element, because:

order of permutation in S_5 is determined by the least common multiplier of disjoint cycle decompositions, but $\operatorname{lcm} 5 = 5$,

$\operatorname{lcm}(4,1) = 4$, $\operatorname{lcm}(3,2) = 6$, $\operatorname{lcm}(2,1,1,1) = 2$, $\cdots$

the lcm cannot be 15, so it contradicts.

[2018. Fall.

7. Consider the identity $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$.

a). Determine the sequence $(C_n) \subseteq \mathbb{R}$ and the range of $x \in \mathbb{R}$ for the identity hold.

b). Show that for each $n \geq 0$,

$$\sum_{k=0}^{n} \frac{C_k}{(n-k)!} = 1.$$

Proof. a).

Notice that $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$ for all $x \in \mathbb{R}$ and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for all } |x|<1.$$

Therefore, the coefficient C_n is determined by the Cauchy Product

$$C_n = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^{n} \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in $\mathbb{R} \cap (-1,1) = (-1,1)$.] 5m.

b). Since

$$\frac{1}{1-x} = e^{x} \cdot \sum_{n=0}^{\infty} C_n x^n,$$ so we can write for $x \in (-1,1)$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^{n} \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each $n \geq 0$, because these are infinitely differentiable, so the coefficients of power series are determined uniquely, by the taylor Theorem. Thus,

$$1 = \sum_{k=0}^{n} \frac{C_k}{(n-k)!} \quad \text{for all } n \geq 0.$$] 4m,

「2019, Spring

6. For each $n \in \mathbb{N}$, let
$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that $(f_n)_{n\geq 1}$ does not converges uniformly.

b) For each fixed $q>1$, Show that (f_n) Converges uniformly on $\{x\in\mathbb{R} \mid |x|\geq q\}$ and $\{x\in\mathbb{R} \mid |x| \leq \frac{1}{q}\}$

a). Given $x \in \mathbb{R}$,

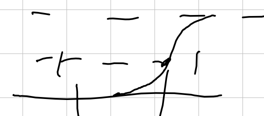
if $|x|<1$, then $f_n(x) \to 0$ as $n\to\infty$, since $|x|^n \to 0$.

if $|x|=1$, then $f_n(x) = \frac{1}{2}$ for all $n \in \mathbb{N}$, since $|x|^n = 1$.

if $|x|>1$, then $f_n(x) = \dfrac{1}{1/|x|^n + 1} \to 1$ as $n\to\infty$, since $1/|x|^n \to 0$ as $n\to\infty$.

Therefore, $f_n \to f$ pointwise where

$$f(x) = \begin{cases} 0 & |x|<1 \\ \frac{1}{2} & |x|=1 \\ 1 & |x|>1. \end{cases}$$



But, for $\epsilon = 1/4$, and for all $n \in \mathbb{N}$,
$$\sup_{0\leq x<1} |f_n(x) - f(x)| = \sup_{0\leq x<1} |f_n(x)| = \sup_{0\leq x<1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$
because given $n\in\mathbb{N}$, we can take $x\in[0,1)$ such that $\dfrac{x^n}{1+x^n} \geq \dfrac{1}{4}$

Since $\displaystyle\lim_{x\to 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$. So, $\displaystyle\sup_{x\in\mathbb{R}} |f_n - f| \geq \sup_{0\leq x<1} |f_n - f|$ implies

$f_n \to f$ but not uniformly $\lrcorner$ lom.

b). If $|x| \geq q > 1$, then $f(x) = 1$. So, enough to show that $\sup|f_n(x) - f(x)| \to 0$.

Since $\displaystyle\sup_{|x|\geq q} |f_n(x) - f(x)| = \sup_{|x|\geq q} \left|\frac{1}{1+|x|^n}\right| \leq \frac{1}{1+|q|^n} \to 0$ as $n\to\infty$, so $f_n \to f$ uniformly

on $|x|\geq q$. Similarly, for $|x| \leq \frac{1}{q}$, $f(x)=0$ and so
$$\sup_{|x|\leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x|\leq\frac{1}{q}} \left|\frac{|x|^n}{1+|x|^n}\right| \leq \frac{1}{|q|^n+1} \to 0 \text{ as } n\to\infty,$$

So proved all cases. $\lrcorner$ lom. $\square$

「2019, Spring.

17. Let $S=\{(x,y,z)\in\mathbb{R}^3 \mid z=x\cdot f(\frac{y}{x}),\ x\neq 0\}$ where $f:\mathbb{R}\to\mathbb{R}$ is differentiable. Show that the tangent plane at any point of S passes through the origin.

Proof By the definition of S, given $(x_0,y_0,z_0)\in S$, a map

$$X: \mathcal{O}\to S: (u,v)\mapsto (u,v,u\cdot f(\tfrac{v}{u}))$$ where $\mathcal{O}$ is an open set containing (x_0,y_0).

is a parametrization of S at (x_0,y_0,z_0). The differential of X at (x_0,y_0) is:

$$dX=\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u})-\frac{v}{u}\cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix}\Bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of S at (x_0,y_0) is generated by

$$\left\{\begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0})-\frac{y_0}{x_0}\cdot f'(\frac{y_0}{x_0})\end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0})\end{pmatrix}\right\}.$$

Now, for $t_1,t_2\in\mathbb{R}$, the coordinate of a point of the tangent plane can be described by

$$(x_0,y_0,x_0\cdot f(\tfrac{y_0}{x_0}))+(t_1,0,t_1 f(\tfrac{y_0}{x_0})-t_1\cdot\tfrac{y_0}{x_0}f'(\tfrac{y_0}{x_0}))$$
$$+(0,t_2,t_2\cdot f'(\tfrac{y_0}{x_0}))$$

is zero when $t_1=-x_0$ and $t_2=-y_0$, so proved. ☒

「2019, Fall.

8. Let $(b_n)_{n\in\mathbb{N}}$ be a decreasing sequence of positive real numbers with $b_n \to 0$. Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of complex numbers such that $\left(\left|\sum_{n=1}^{N} a_n\right|\right)_{N\in\mathbb{N}}$ is bounded sequence.

a). Show that $\sum_{n=1}^{\infty} a_n b_n$ converges.

b). Show $\sum_{n=2}^{\infty} \frac{\sin(nx)}{\log n}$ converges for all $x\in[-\pi,\pi]$.

Proof. Denote $A_n = \sum_{k=1}^{n} a_k$ and $A_0=0$. And let $M>0$ be a bound for $\sum a_n$.

$$\sum_{n=1}^{N} a_n b_n = \sum_{n=1}^{N}[A_n - A_{n-1}]\cdot b_n = \sum_{n=1}^{N} A_n b_n - \sum_{n=1}^{N} A_{n-1}\cdot b_n \quad \left(\sum_{n=1}^{N} A_{n-1} b_n = \sum_{n=0}^{N-1} A_n b_{n+1}\right)$$

$$= \sum_{n=1}^{N} A_n b_n - \left(\sum_{n=1}^{N} A_n \cdot b_{n+1} + A_0 b_1 - A_N b_{N+1}\right)$$

$$= \sum_{n=1}^{N} A_n (b_n - b_{n+1}) + A_N b_{N+1}.$$

Therefore, for any $m > n \geq 1$,

$$\sum_{k=n}^{m} a_k b_k = \sum_{k=n}^{m} A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_{n-1} b_n$$

Now, using the hypothesis,

$$\left|\sum_{k=n}^{m} a_k b_k\right| \leq \sum_{k=n}^{m} |A_k|\cdot|b_k - b_{k+1}| + |A_m||b_{m+1}| + |A_{n-1}|\cdot|b_n|$$

$$\leq M\cdot\left(\sum_{k=n}^{m} |b_k - b_{k+1}| + |b_{m+1}| + |b_n|\right)$$

$$= M\cdot\left(\sum_{k=n}^{m} (b_k - b_{k+1}) + b_{m+1} + b_n\right)$$

$$= M\cdot 2b_n \to 0 \text{ as } n\to\infty, \text{ so proved.}$$

b). Since $1/\log n$ is decreasing and positive real numbers, so by a), enough to show that $\left|\sum \sin(nx)\right|$ is bounded for all $x\in[-\pi,\pi]$.

observe that: $2\sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

Therefore, $2\sin x \cdot \sum_{n=1}^{N} \sin(nx) = \sum_{n=1}^{N} \cos[(n-1)x] - \cos[(n+1)x] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x| \leq 4$

So for $x\in[-\pi,\pi]\setminus\{0,\pi,-\pi\}$, $\left|\sum_{n=2}^{N} \sin nx\right| \leq \frac{2}{|\sin x|}$ for any $N\in\mathbb{N}$, bounded.

for $x = 0$ or $\pm\pi$, then the sum is just zero. ∎

[2019, Fall,

9. Let $\varphi: [0,1] \to \mathbb{R}$ be a continuous map. Evaluate

$$\lim_{\varepsilon \to 0} \int_0^1 \left[\frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

「2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of $y'' - 3y' + 2y = 0$ is $r^2 - 3r + 2 = (r-1)(r-2)$, therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^{t} + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution $y_p(t)$. First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$y_p(x) = \int_{x_0}^{x} \begin{vmatrix} e^t & e^{x} \\ e^{2t} & e^{2x} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} \, dt$$

$$= \int_{x_0}^{x} \frac{e^{t+2x} - e^{2t+x}}{e^{3t} + e^{2t}} \, dt$$

$$= e^{2x} \int_{x_0}^{x} \frac{1}{e^{2t} + e^{t}} \, dt - e^{x} \int_{x_0}^{x} \frac{1}{e^{t}+1} \, dt$$

Meanwhile, since $\displaystyle\int \frac{1}{e^t+1} \, dt = \int \frac{e^{-t}}{1+e^{-t}} \, dt = -\log(1+e^{-x})$

and $\displaystyle\int \frac{1}{e^{2t}+e^{t}} \, dt = \int \frac{e^{-t}}{e^{t}+1} \, dt = \int \frac{-u}{\frac{1}{u}+1} \frac{1}{u} \, du = \int \frac{-u}{1+u} \, du$

$$= \int -1 + \frac{1}{1+u} \, du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$y_p + C_2 \varphi_1 + C_3 \varphi_3 = \left[-e^{x} + e^{2x} \log(1+e^{-x}) + e^{x} \cdot \log(1+e^{-x})\right]$$

$$+ C_2 \cdot e^{x} + C_3 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R},$$

[2020, Fall.

1. Let p be a Prime Find the number of subgroups H of $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$, $|H| = p^2$

Proof. By the fundamental theorem of finitely generated abelian group, either $H \cong \mathbb{Z}_{p^2}$ or $\mathbb{Z}_p \times \mathbb{Z}_p$.]

If $H \cong \mathbb{Z}_p \times \mathbb{Z}_p$, then for all $x \in H$, $p \cdot x = 0$. Therefore, $H \subseteq \{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\}$.

Meanwhile, $pa = 0$ and $pb = 0$ if and only if $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$ and $b \in \langle p^2 \rangle \subseteq \mathbb{Z}_{p^3}$, so $\{(a,b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a,b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$.

Since $\langle p \rangle \times \langle p^2 \rangle$ has already order p^2, so $H = \langle p \rangle \times \langle p^2 \rangle$.]

If $H \cong \mathbb{Z}_{p^2}$, that is, H is generated by order p^2 element,

Since $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ has order $p^2 \iff p^2 \cdot x = 0$ and $px \neq 0$.

Every element $a \in \mathbb{Z}_{p^2}$ satisfies $p^2 a = 0$, and $b \in \mathbb{Z}_{p^3}$ satisfies $p^2 \cdot b = 0$ if and only if $b \in p\mathbb{Z}_{p^3}$ and $\mathbb{Z}_{p^3}$ has number of p^2 of p-multiples, so there are $p^2 \cdot p^2 = p^4$ elements satisfying $p^2 \cdot x = 0$.

And, $px = 0$ if and only if $a \in p\mathbb{Z}_{p^2}$ and $b \in p^2\mathbb{Z}_{p^3}$, where $x = (a,b)$.

So, there are p^2 elements satisfying $px = 0$.

Consequently, there are $x^4 - x^2$ elements x such that $H = \langle x \rangle$.

Moreover, the cyclic group of order p^2 has $\varphi(p^2) = p^2 - p$ generators, so the possible $H \cong \mathbb{Z}_{p^2}$ are

$$(p^4 - p^2)/(p^2 - p)$$

numbers.

Finally, the number of H is $1 + (p^4 - p^2)/(p^2 - p)$ ☒

Note: for a prime p, $\varphi(p^n) = p^n - p^{n-1}$.

「2020 Fall.

2. For each $n \in \mathbb{N}$, let $a_n = \cos(2\pi/n)$.

a). Prove that a_n is algebraic over $\mathbb{Q}$

b). Calculate $\deg_{\mathbb{Q}}(a_5)$.

Proof. a). Let $\xi = \exp(2\pi i/n)$. Then, $\xi^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$ gives that $a_n = \cos(2\pi/n) = (\xi + \xi^{-1})/2$.

Meanwhile, since $\xi^n = \exp 2\pi i = 1$, so $\xi^n - 1$ gives ξ is algebraic over $\mathbb{Q}$.

Therefore, ξ^{-1} is also algebraic, thus $a_n = \dfrac{\xi + \xi^{-1}}{2}$ is algebraic over $\mathbb{Q}$.

b). Let $w = \exp(2\pi i/5)$. Since w is the primitive 5th root of unity, $w^4 + w^3 + w^2 + w + 1 = 0$. And, dividing by w^2 then $2a_5 = w + w^{-1}$ implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$
$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$
$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since $4x^2 + 2x - 1$ is irreducible over $\mathbb{Q}$ by the discriminant $2^2 + 16 = 20$ but $\sqrt{20} \notin \mathbb{Q}$, so $\deg_{\mathbb{Q}}(a_5) = 2$. $\blacksquare$

「Fail.

a) Notice that: De Moivre's formular gives

$$e^{i \cdot nz} = (\cos nz + i \cdot \sin nz) = (\cos z + i \cdot \sin z)^n$$

Therefore

$$\cos n \cdot \frac{2\pi}{n} = 1 = \mathrm{Re}\left(\cos \frac{2\pi}{n} + i \cdot \sin \frac{2\pi}{n}\right)^n$$
$$= \mathrm{Re}\left(\sum_{k=0}^{n} \left(\cos \frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n}\right)^{n-k} \cdot \binom{n}{k}\right)$$

Meanwhile, $\sin\left(\frac{2\pi}{n}\right)^2 = 1 - \cos\left(\frac{2\pi}{n}\right)^2$ and the real part of $\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos \frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n}\right)^{n-k}$ Yield only $n-k$ is even, since i^{n-k} is real $\Leftrightarrow n-k$, so $\mathrm{Re}\left(\sum_{k=0}^{n} \binom{n}{k} \cdot \left(\cos \frac{2\pi}{n}\right)^k \cdot \left(i \cdot \sin \frac{2\pi}{n}\right)^{n-k}\right) - 1 = 0$ is evaluated $\cos\left(\frac{2\pi}{n}\right)$ for some polynomial over $\mathbb{Q}$, so a_n is algebraic over $\mathbb{Q}$.

b). From above argument,

$$1 = \cancel{\underset{k=0}{1 \cdot a_5^0 \cdot 0}} + \underset{k=1}{5 \cdot a_5^1 \cdot (1 - a_5^2)} - \underset{k=3}{10 \cdot a_5^3 \cdot (1 - a_5^2)^2} + \underset{k=5}{a_5^5}$$
$$= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5$$
$$= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7.$$

So, $10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0$.

「2020. Fall

3. Let $\mathbb{K}$ be a field. Show that $(x^3-y^5) \subseteq \mathbb{K}[x,y]$ is prime ideal.

Proof. Since $\mathbb{K}[x,y]$ is U.F.D., every irreducible element is prime.
Thus, enough to show that x^3-y^5 is irreducible.
Notice that: $\mathbb{K}[x,y] = \mathbb{K}[y][x]$ by definition, So $x^3-y^5 \in \mathbb{K}[y][x]$.
Since $\gcd(1, y^5) = 1$, x^3-y^5 is primitive, by the Gauss Lemma,
x^3-y^5 is irreducible over $\mathbb{K}[y] \iff x^3-y^5$ is irreducible over $\mathbb{K}(y)$.
But, since x^3-y^5 has degree 3, x^3-y^5 is reducible only if it has a linear factor, in other say, x^3-y^5 has a root in $\mathbb{K}(y)$.
Suppose that for some $f(y)/g(y) \in \mathbb{K}(y)$,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then, $f(y)^3 = y^5 \cdot g(y)^3$, But $\deg f(y)^3 \equiv 0 \mod 3$, and

$$\deg y^5 \cdot g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \mod 3.$$

Therefore, there is no such a root, so x^3-y^5 is irreducible over $\mathbb{K}[y]$,
So it is irreducible over $\mathbb{K}[x,y]$.

Note:

$$K[x,y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

「Modified

Prove that $\gcd(m,n)=1$ implies $x^m - y^n \in F[x,y]$ is irreducible.

「2020. Fall.

4. Let $W \subseteq V = \mathbb{R}^n$ be a subspace. Show that there exists a homogeneous system of linear equation whose solution space is W.

Proof. Let $\beta = \{\alpha_1, \cdots, \alpha_k\}$ be a basis for W.

Then, we can extend the basis β to $\overline{\beta} = \beta \cup \{\alpha_{k+1}, \cdots, \alpha_n\}$ for V.

Now, consider the map:

$$f_j : V \to \mathbb{R} ; \sum c_i \alpha_i \mapsto c_j$$

Then, $W = \ker f_{k+1} \cap \cdots \cap \ker f_n$, so for the standard basis S for V,

$$A = \begin{pmatrix} [f_{k+1}]_S \\ \vdots \\ [f_n]_S \end{pmatrix}$$ satisfies $\ker A = W$, so

($[f]_S$ is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$ is the system we desired.

「2021. Spring.

1. Show that no group of 2020 is simple.

proof $2020 = 2^2 \cdot 5 \cdot 101$. Denote n_p as the number of Sylow p-subgroups.
By the Sylow theorem, $n_{101} \equiv 1 \mod 101$ and $n_{101} \mid 20$, so n_{101} must be 1.
Therefore, the Sylow 101-subgroup is normal, so proved.

「Without using Sylow theorem.
By the Cauchy's theorem, G has cyclic subgroup $H \leq G$ of order 101.
Denote $A = \{gHg^{-1} \mid g \in G\}$. Then, A is orbit of H from the action

$$G \times \{S \leq G\} \to \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of H is $G_H = N_G(H)$, we obtain

$$\# A = |G : N_G(H)|.$$

And since $H \leq N_G(H)$,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \# A \leq 20.$$

Now, consider the action

$$H \times A \to A : (h, K) \mapsto hKh^{-1}.$$

Given $K \in A$, the H-orbit of this action is $\mathrm{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$ and

$$\# \mathrm{Orb}_H(K) = |H : \mathrm{Stab}_H(K)|, \ \mathrm{Stab}_H(K) \leq H \Rightarrow |\mathrm{Stab}_H(K)| \in \{1, 101\}.$$

So, $\# \mathrm{Orb}_H(K) \in \{1, 101\}$ and $\mathrm{Orb}_H(K) \leq A$, so $\# \mathrm{Orb}_H(K) = 1$.
therefore, every element in H normalizes $K \in A$, that is, $H \leq N_G(K)$.
Now, for some $K \in A$, if $H \neq K$ then $HK \leq G$ and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since $101^2 \nmid 2020$, so $A = \{H\}$. That is, $H \trianglelefteq G$. ☒

「2021. spring

17. Let f be a polynomial given by

$$f(z) = a_0 + a_1 z + \cdots + a_d z^d \quad \text{over } \mathbb{C}, \; a_d \neq 0.$$

$\boxed{8m.}$

Evaluate the value $\displaystyle \lim_{R\to\infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz$

Proof. By the fundamental theorem of Algebra, f has d-roots $z_1, \cdots, z_d \in \mathbb{C}$.

That is, $f(z) = a_d (z - z_1) \cdots (z - z_d)$

Moreover, $\dfrac{f'(z)}{f(z)} = \dfrac{1}{z - z_1} + \cdots + \dfrac{1}{z - z_d}$, just calculation.

Therefore, for $R > \max\{|z_1|, \cdots, |z_d|\}$, by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)}\, dz = \sum_{n=1}^{d} \operatorname{Res}\left(\frac{z}{z - z_n}, z_n \right) = \sum_{n=1}^{d} \lim_{z\to z_n} z = z_1 + \cdots + z_n$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \cdots + z_n) = 2\pi i \cdot \left(\frac{-a_{d-1}}{a_d} \right),$$

$\boxtimes$

「2022. Spring.

8. Define $$f : \mathbb{R} \to \mathbb{R} ; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2+n^2}$$

Then, f is differentiable on $\mathbb{R}$.

proof. For each $m \in \mathbb{N}$, define
$$f_m(x) = \sum_{n=1}^{m} \frac{1}{x^2+n^2}.$$

At first, $f_m(0) \to f(0)$ as $m \to \infty$, because that
$$\sum_{n=2}^{m} \frac{1}{n^2} < \sum_{n=2}^{m} \frac{1}{n^2-n} = \sum_{n=2}^{m} \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \to 1 \text{ as } m \to \infty,$$
so the increasing bounded sequence $f_m(0)$ converges. Now,
$$f_m'(x) = \sum_{n=1}^{m} - \frac{2x}{(x^2+n^2)^2}$$
converges uniformly, because

if $|x| \le 1$, then
$$\left| \frac{-2x}{(x^2+n^2)^2} \right| \le \frac{2}{n^4} \quad \text{and} \quad \sum \frac{1}{n^4} < \infty,$$
if $|x| > 1$, then
$$\left| \frac{-2x}{(x^2+n^2)^2} \right| \le 2 \cdot \left| \frac{x}{x^2+n^2} \cdot \frac{1}{x^2+n^2} \right| \le 2 \cdot \frac{1}{n^2} \quad \text{and} \quad \sum \frac{1}{n^2} < \infty$$
therefore, the Weierstrass M-test gives f_m' converges uniformly.

Now, f_m converges at some point and f_m' converges uniformly on $\mathbb{R}$, the limit function f is differentiable. 」8m.

First, for any $x \in \mathbb{R}$, $|x^2+n^2| \ge n^2$, so
$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2+n^2} \right| \le \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^{m} \frac{1}{n^2} < \sum_{n=2}^{m} \frac{1}{n^2-n} = \sum_{n=2}^{m} \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \to 1,$$
as $m \to \infty$. Therefore, given $\varepsilon > 0$, for some $N \in \mathbb{N}$, $m > n \ge N$ implies
$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^{m} \frac{1}{x^2+k^2} \right| \le \sup_{x \in \mathbb{R}} \sum_{k=n}^{m} \left| \frac{1}{x^2+k^2} \right| < \sum_{k=n}^{m} \frac{1}{k^2} < \varepsilon.$$

Now, define for each $m \in \mathbb{N}$,
$$f_m(x) = \sum_{n=1}^{m} \frac{1}{x^2+n^2}.$$

We already shown $f_m \to f$ uniformly. And,
$$f_m'(x) = \sum_{n=1}^{m} - \frac{2x}{(x^2+n^2)^2}$$
Converges uniformly, because

「2022. Fall.

1. Let G be a group and H and K are normal subgroup of G. If $H \cap K = \{e\}$, then $HK \cong H \times K$.

Proof. Define a map

$$\varphi : H \times K \to HK : (h, k) \mapsto hk.$$

It is well-defined by the definition of HK, and it is homomorphism; given $h_1, h_2 \in H$ and $k_1, k_2 \in K$,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 k_1 k_2$$

meanwhile, given $h \in H$ and $k \in K$, $hkh^{-1}k^{-1} \in K$ and $\in H$, because $hkh^{-1} \in K$ and $kh^{-1}k^{-1} \in H$, by the normality. But, $H \cap K = \{e\}$ implies $hkh^{-1}k^{-1} = e$, so $hk = kh$, these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1)) \varphi((h_2, k_2)),$$ so φ preserves the operation.

And, $\varphi[H \times K] = HK$ by just definition, φ is onto. Now, the first isomorphism theorem gives that $H \times K / \ker \varphi \cong HK$.

And, $$\ker \varphi = \{(h, k) \in H \times K \mid hk = e\} = \{e\},$$

so $H \times K \cong H \times K / \{e\}$, it proves the assertion. ▣

2022. Fall.

8. For Continuous functions $g:[0,1]\to\mathbb{R}$ and $h:[-1,1]\to\mathbb{R}$, define

$$f:[0,1]\to\mathbb{R}: x\mapsto \int_0^1 h(x-y)g(y)\,dy$$

For each $n\in\mathbb{N}$, define

$$f_n:[0,1]\to\mathbb{R}: x\mapsto \frac{1}{n}\cdot\sum_{i=0}^{n-1} h\left(x-\frac{i}{n}\right)\cdot g\left(\frac{i}{n}\right)$$

Prove that $f_n\to f$ uniformly on $[0,1]$

Proof. Define $H:[0,1]\times[0,1]: (x,y)\mapsto h(x-y)\cdot g(y)$.

Since h and g both are continuous, therefore H is continuous and since $[0,1]^2$ is compact set, so H is continuous uniformly.

Given $\varepsilon>0$, there exists $\delta>0$ such that for any $(x_1,y_1),(x_2,y_2)\in[0,1]^2$,

$$\|(x_1,y_1)-(x_2,y_2)\|<\delta \Rightarrow \|H(x_1,y_1)-H(x_2,y_2)\|<\varepsilon.$$

Now, given $x\in[0,1]$, and $n>\frac{1}{\delta}$, ($n>\frac{1}{\delta}$ gives $\frac{1}{n}<\delta$)

$$\begin{aligned}
|f(x)-f_n(x)| &= \left|\int_0^1 H(x,y)\,dy - f_n(x)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \frac{1}{n}\cdot\sum_{i=0}^{n-1} H\left(x,\frac{i}{n}\right)\right| \\
&= \left|\sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H(x,y)\,dy - \sum_{i=1}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} H\left(x,\frac{i}{n}\right)dy\right| \\
&\leq \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} \left|H(x,y)-H\left(x,\frac{i}{n}\right)\right|dy \\
&< \sum_{i=0}^{n-1}\int_{\frac{i}{n}}^{\frac{i+1}{n}} \varepsilon\,dy = \varepsilon.
\end{aligned}$$

Therefore, $n>\frac{1}{\delta} \Rightarrow \sup_{x\in[0,1]}|f_n(x)-f(x)|<\varepsilon$, So, proved. $\square$

┌2022. Fall.

13. Let S be the subset of $\mathbb{R}^3$ given by $\{(x,y,z) \mid z^2 = x^2+y^2,\ z \geq 0\}$.
Show that S is not regular surface.

prof. Suppose that S is a regular surface. Then, for $(0,0,0) \in S$, there exists a neighbor V containing $(0,0,0)$ such that $V \cap S$ is a graph of

$$z = f(x,y) \quad \text{or} \quad y = g(x,z) \quad \text{or} \quad z = h(y,z)$$

where f, g, h are differentiable. Since projection of S into xz, yz-plain are not 1-1, g, h cannot be such a map.
Meanwhile, $f(x,y) = \sqrt{x^2+y^2}$ at locally $(0,0,0)$, but f is not differentiable at $(x,y) = (0,0)$, so it cannot be possible. Therefore S is not regular.

(More specifically, $(0, \pm\varepsilon, \sqrt{1-\varepsilon^2}) \in S$ for any $\varepsilon > 0$, so there is no 1-1 yz-projection. Similarly xz.).

「2023. Spring

4. Let $V = \{id, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$,

a). Show that V is normal subgroup.

b). $S_4/V \cong S_3$,

☐ 2023, Spring.

7. Let $f: \mathbb{C} \to \mathbb{C}$ be an entire satisfying 10m

$$u^2(z) - 2\cdot v^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that f is constant.

Proof. observe that ~~differentiating gives~~ that

$$\cancel{2uu_x - 2vv_x = 0 \Rightarrow u\cdot u_x = vv_x = -v\cdot u_y}$$

Observe that: $f(z) = 0 \Leftrightarrow u(z) = v(z) = 0$

But, $u^2 - 2v^2 = 1$, so $f(z) \neq 0$ for all $z \in \mathbb{C}$. Now,

$$\frac{1}{f^2(z)} = \frac{1}{u^2(z) + v^2(z)} = \frac{1}{1 + 3v^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function $\frac{1}{f^2}$ is constant, so f is also constant function. ☑

Or, $|f^2| = u^2 + v^2 = 1 + 3v^2 \Rightarrow \frac{1}{f} \leq \frac{1}{\sqrt{1+3v^2}} \leq 1$.

「2024, Spring.

1. Let $A, B \in M_{m\times n}(F)$. Show that $\text{rank}(A+B) \le \text{rank}\, A + \text{rank}\, B$. $\boxed{5m}$

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank}\, A = \dim \text{Im}\, A, \quad \text{rank}\, B = \dim \text{Im}\, B.$$

Given $\alpha \in \text{Im}(A+B)$, $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im}\, A + \text{Im}\, B$ for some $\beta \in F^m$, therefore $\text{Im}(A+B) \subseteq \text{Im}\, A + \text{Im}\, B$. Now,

$$\text{rank}(A+B) \le \dim(\text{Im}A + \text{Im}\, B) \le \dim \text{Im} A + \dim \text{Im}\, B = \text{rank}\, A + \text{rank}\, B.$$

2. Suppose that P and Q are positive definite such that $P^2 = Q^2$. Show $P = Q$.

Proof. Since P is Hermitian, there exists an orthonormal basis $\mathcal{B} = \{\alpha_1, \cdots, \alpha_n\}$ such that $P(\alpha_i) = c_i\alpha_i$ for some $c_i > 0$, for each i.

Let $1 \le i \le n$ be fixed. Then, by the assumption, $\boxed{15m}$

$$P^2\alpha_i = c_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - c_i^2 I)\alpha_i = (Q - c_i I)(Q + c_i I)\alpha_i = 0.$$

Since Q is positive definite, $(Q - c_i I)\alpha_i = 0$, because; if it is not zero, then

$$(Q - c_i I)(Q + c_i I)\alpha_i = (Q + c_i I)(Q - c_i I)\alpha_i = 0,$$

So $-c_i$ is negative eigenvalue of Q, it contradicts to Q is positive definite. Therefore, for each $1 \le i \le n$,

$$P\alpha_i = c_i\alpha_i = Q\alpha_i,$$

and since $\mathcal{B}$ is basis, therefore $P = Q$. ∎

「2024, Spring

3. Let $T: \mathbb{R}^5 \to \mathbb{R}^5 ; x \mapsto Ax$ where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists T-invariant subspace $V \subseteq \mathbb{R}^5$ s.t $\dim V = 4$?

b). 〃 s.t $\dim V = 2$, $T|_V$ is diagonalizable?

c) 〃 $\dim V = 3$, 〃

Proof.

a). Let $V = \langle e_1, e_2, e_3, e_4 \rangle$. Then,

$$Ae_1 = e_1, \quad Ae_2 = e_1 + e_2, \quad Ae_3 = e_3, \quad Ae_4 = e_3 + e_4$$

are all contained in V, so $T[V] \subseteq V$. (clearly $\dim V = 4$)

b). Let $V = \langle e_1, e_3 \rangle$. Then $Ae_1 = e_1$, $Ae_3 = e_3$ are contained in V, and

$$[T|_V]_{\{e_1, e_3\}} = I_2.$$ So $T|_V$ is diagonalizable.

c). Observe characteristic polynomial of T is $(t-1)^5$.
If such a subspace V exists, then the characterstic polynomial of $T|_V$ is $(t-1)^3$. Since $T|_V$ is diagonalizable if and only the minimal polynomial splits and separable, thus the minimal polynomial of $T|_V$ must be $(t-1)$. That is, $T|_V = I_V$ but it implies that $\ker(T-I)$ has dimension more than or equal 3, it contradicts to

$$\dim \ker(A - I) = \dim \ker \begin{pmatrix} 0 & 1 & & & \\ & 0 & 0 & 0 & \\ & & 0 & 1 & \\ & & & 0 & 1 \\ 0 & & & & 0 \end{pmatrix} = 2.$$

「2024, Spring.

5. Let K be a field, and $L_1, L_2 \subseteq \overline{K}$ be finite extensions of K. 41

a). Prove $L = L_1L_2$ is finite extension.

b). If L_1 and L_2 are Galois over K, then L/K is Galois.

a). Since L_1 and L_2 both are finite, denote

$L_1 = K(\alpha_1, \cdots, \alpha_n)$ and $L_2 = K(\beta_1, \cdots, \beta_r)$ where $\alpha_i, \beta_j \in \overline{K}$ are algebraic over K.

Now, $$L = L_1L_2 = \bigcap_{\substack{E \leq \overline{K} \\ L_1, L_2 \leq E}} E = K(\alpha_1, \cdots, \alpha_n, \beta_1, \cdots, \beta_r)$$

is finitely generated by algebraic elements α_i, β_j, so L is finite. ⅃4m.

~~b). Since L_1 and L_2 are splitting field of K, let f_1 and f_2 be~~
~~polynomials over K and such that L_i is the splitting field of f_i, $i = 1, 2$.~~
~~It is possible, because L_1, L_2 both are finite extension.~~
~~Now let E be the splitting field of f_1f_2. Then,~~
~~$E \subseteq L_1L_2$, because E is the smallest field containing all roots of f_1f_2,~~
~~$L_1L_2 \subseteq E$ because E contains all roots of f_1 and f_2. So, $L = E_1E_2$,~~
~~therefore L is splitting field over K. ⅃7m.~~

Since L_1 and L_2 both are Galois, for some separable f_1 and f_2 over K, L_1 and L_2 are splitting field of f_1 and f_2 respectively.

L_1L_2 가 f_1f_2 의 splitting field => ok

근데 f_1f_2가 separable 이란 보장 X.

중복 linear factor 소거 => f_1f_2가 여전히 over K 인가?

Revised in the Algebra/Field Theory/

「2024 spring.

8. Let $C_b(\mathbb{R})$ be the set of all real-valued bounded continuous functions on $\mathbb{R}$. For two functions $f, g \in C_b(\mathbb{R})$, define

$$d(f,g) := \sup_{x\in\mathbb{R}} |f(x) - g(x)|.$$

Let $\{f_n\}_{n=1}^{\infty}$ be a sequence in $C_b(\mathbb{R})$ such that:

$\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$ such that $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$.

a). Prove that there exists $f \in C_b(\mathbb{R})$ such that $\lim_{n\to\infty} d(f, f_n) = 0$

b). Prove that $\lim_{n\to\infty}\left(\sup_{x\in\mathbb{R}}|f_n(x)|\right) = \sup_{x\in\mathbb{R}}|f(x)|$.

Proof. For each $x \in \mathbb{R}$, a sequence $\{f_n(x)\}$ in $\mathbb{R}$ is Cauchy sequence by hypothesis, Since $\mathbb{R}$ is complete, $\lim_{n\to\infty} f_n(x)$ exists, Put this $f(x) \in \mathbb{R}$.

a).

Since $x \in \mathbb{R}$ was arbitrary, $f(x)$ is well-defined on $\mathbb{R}$, but still not necessarily $f \in C_b(\mathbb{R})$, and $d(f, f_n) \to 0$.

Given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $m > n \geq N$ implies that $\sup_{x\in\mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$. So, for any $x \in \mathbb{R}$, $|f_m(x) - f_n(x)| \leq \varepsilon$. Letting $m \to \infty$, we obtain $|f(x) - f_n(x)| \leq \varepsilon$. Since x was arbitrary, $\sup_{x\in\mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$, for $n \geq N$.

Therefore, the triangle inequality gives $|f(x)| \leq |f_n(x)| + \varepsilon$ for all $x \in \mathbb{R}$ and since f_n was bounded, f is bounded. And, the argument above proves $\sup_{x\in\mathbb{R}} |f - f_n| \to 0$ as $n \to \infty$, So since f_n is continuous and

$f_n \to f$ uniformly, so f is also continuous, so $f \in C_b(\mathbb{R})$ and a) is proved.

b). Since $\left|\sup_{x\in\mathbb{R}}|f_n(x)| - \sup_{x\in\mathbb{R}}|f(x)|\right| \leq \sup_{x\in\mathbb{R}}|f_n(x) - f(x)| \to 0$ as $n \to \infty$,

Proved directly.

「2025 Spring.

13. Let D^2 be a closed unit disc in $\mathbb{R}^2$ centered at the origin.

a). Show that the quotient space $\mathbb{R}^2/D^2$ is homeomorphic to $\mathbb{R}^2$

b). Let $U = \text{int } D$. Determine $\mathbb{R}^2/U \cong \mathbb{R}^2$.

Proof. Let $\sim$ be a relation defined on $\mathbb{R}^2$ as:

$$(x_1, y_1) \sim (x_2, y_2) \iff \{(x_1, y_1), (x_2, y_2)\} \subseteq D^2.$$

Then, it is clearly equivalent relation, and $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$, by definition.

a). Define a map

$$f: \mathbb{R}^2 \to \mathbb{R}^2 : (x, y) \mapsto$$

Then, given $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$,

$$(x_1, y_1) \sim (x_2, y_2) \Leftarrow$$

[2025. Spring.

14. For $a, b, c > 0$, Consider the ellipsoid

$$E_{a,b,c} = \left\{ (x,y,z) \in \mathbb{R}^3 \;\middle|\; \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

a). If $a=b$, then $E_{a,a,c}$ is a surface of revolution (Prove.)

b) Give a parametrization of $E_{a,a,c}$. Compute the first fundamental form of this parametrization and the Christoffel symbol.

c). Show that the curve $E_{a,b,c} \cap \{x=0\}$ is the image of a geodesic.

a). Let

$X : (0,2\pi) \times (0,2\pi) \to \mathbb{R}^3 : (u,v) \mapsto (a \cdot \cos v \cdot \cos u,\ a \cdot \cos v \cdot \sin u,\ c \cdot \sin v).$

「2025. Fall

$\boxed{8m}$

1. a). Show that $\phi : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p'(0)$ is linear functional.

b). Find a dual basis corresponding to $\{1, t, \cdots, t^n\}$ of $P_n(\mathbb{R})$.

Proof. Let $f(x) = \sum_{i=0}^{n} a_i x^i$ and $g(x) = \sum_{i=0}^{n} b_i x^i$ and $c \in \mathbb{R}$ be given.

Then, $\phi(cf+g) = (cf+g)'|_{t=0} = c \cdot f'|_{t=0} + g'|_{t=0} = c \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a).

To show b), let

$$\beta^* := \left\{ \frac{1}{k!} \cdot \phi_k : P_n(\mathbb{R}) \to \mathbb{R} : p(t) \mapsto p^{(k)}(0) \;\middle|\; k = 0, 1, \cdots, n \right\}$$

Then, for each $k = 0, 1, \cdots, n$,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in β^* are all linear functional, so the β^* is dual basis.

「2025. Fall

2. Let $A \in M_{m\times m}(\mathbb{R})$, $\boxed{15m}$: A^n 이라 얼탐, $n=2$ 인때부터 생각

a). For $n \in \mathbb{N}$, $\operatorname{rank} A^n \le \operatorname{rank} A$.

b). Let $L_A : \mathbb{R}^m \to \mathbb{R}^m ; x \mapsto Ax$.

Show that for $n \ge 2$,

$$\operatorname{rank} A^n = \operatorname{rank} A \text{ implies } \ker L_A \cap \operatorname{Im} L_A = \{0\}.$$

Sol) a). Given $x \in \operatorname{Im} A^n$, $x = A^n y$ for some $y \in \mathbb{R}^m$.
So, $x = A^n y = A^{n-1} Ay \in \operatorname{Im} A$, for $n \ge 2$. Thus $\operatorname{Im} A^n \subseteq \operatorname{Im} A$, it follows $\operatorname{rank} A^n \le \operatorname{rank} A$. If $n=1$, Clear.

b). Given $x \in \ker A$, $A^n x = A^{n-1} Ax = A^{n-1} 0 = 0$, so $x \in \ker A^n$, so $\ker A \subseteq \ker A^n$.
Since $\operatorname{rank} A^n = \operatorname{rank} A$, the dimension theorem gives $\operatorname{nullity} A^n = \operatorname{nullity} A$.
Therefore, since $\mathbb{R}^m$ is finite dimensional, so is $\ker A^n$, thus $\ker A = \ker A^n$.
Now, let $x \in \ker L_A \cap \operatorname{Im} L_A$ be given. Then, $Ax = 0$ and $x = Ay$ for some $y \in \mathbb{R}^m$.
Then, $A^{n-1} \cdot x = A^n y = 0$, because $Ax = 0$, so $A^{n-2} \cdot Ax = 0$, and this implies $y \in \ker A^n$.
Since $\ker A^n = \ker A$, $Ay = 0$, so $x = Ay = 0$, so proved. ⊥

「2025. Fall

3. Find all $a, b, c \in \mathbb{R}$ such that $\boxed{6m}$

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol). Since the Charatorstic polynomial of A is regardless of a, b, c,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the charatorstic polynomial, the A is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

Calculating gives $= \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$

therefore, $a=0$, $b \in \mathbb{R}$, $c \in \mathbb{R}$ are all possible (a, b, c).

「2025. Fall」

20m

4. Let $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be the set of all pairs (A, B) such that

$$\det A = \pm \det B.$$

Show that N is normal and determine the factor group $(G \times G)/N$.

Sol). Let $(A, B) \in N$ and $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$ be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

and $\det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$

so, since $(A, B) \in N$, $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore $(C, D)(A, B)(C, D)^{-1} \in N$. And, to show N is subgroup,

first, $(I, I) \in N$ trivially, so $N \neq \emptyset$. Let $(A, B), (C, D) \in N$ be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$.

therefore, by the subgroup criterion, N is subgroup, and normal. ⊣ 10m.

Define a map

$$\varphi : GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \to \mathbb{R}^\times : (A, B) \mapsto (\det A \cdot \det B^{-1})^2$$

Then, $\ker \varphi = N$ and φ is homomorphism, onto $(0, \infty)$, because the det preserves multiplication, so is φ, and ontoness is given by: fix $B = I$, and $A = \mathrm{diag}(\sqrt{r}, 1, 1, \cdots, 1)$ where $r > 0$ is arbitrary.

By the first-isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times)$$ ⊣ 10m. ⓟ

Comment: φ 을 먼저 찾고 시작하고, hom 임을 보이면 앞 계산 과정이 날라감!

[2025. Fall

10m

5. Let G be an order 21 subgroup of $\mathbb{C}^\times$.

a) G is cyclic.

b). For any $g \in G$, $x^{12} = g^{15}$ has 3 distinct solutions in G.

Proof. a). Since $\mathbb{C}$ is abelian, so G is also abelian.
By the Cauchy's theorem, G has order 3 element x and order 7 element y.
Now, since 3 and 7 are relatively prime, $|xy| = 21$, so $G = \langle xy \rangle$.

b) By a), $G \cong \mathbb{Z}/21\mathbb{Z}$. So, we can write the given equation as

$$12x \equiv 15g \mod 21 \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of g, $\gcd(12, 21) = 3$ divides $15g$, so it has three distinct solutions.

Note: $ax \equiv b \mod n$ has a solution iff $\gcd(a, n) \mid b$.
And in such a case, it has $\gcd(a, n)$ solution

「2025. Fall,

6. Let $\{a_n\}$ be a bounded sequence in $\mathbb{R}$, Define $\boxed{22m}$

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\},$$

a). Prove that $\{a_n\}^*$ is closed.

b). Show $\limsup_{n\to\infty} a_n = \sup\{a_n\}^*$

c). Is there a sequence $\{b_n\}$ such that $\{b_n\}^* = [0,1]$?

Proof a). If $\{a_n\}^*$ is finite, then clearly closed in $\mathbb{R}$.
Assume that $\{a_n\}^*$ is infinite. Since $\{a_n\}$ is bounded, so the set of subsequential limits $\{a_n\}^*$ is bounded set. So it has a limit point.
Let a be a limit point of $\{a_n\}^*$. Then, given $k \in \mathbb{N}$, there exists a subsequential limit p of $\{a_n\}$ such that $p \in B_{\frac{1}{k}}(a) \setminus \{a\}$.
For $k=1$, choose such a $p_1 \in B_1(a)\setminus\{a\}$, and since p is a subsequential limit, choose $a_{p_1} \in B_{1-|p-a|}(p)$, then $a_{p_1} \in B_1(a)$.

For chosen $a_{p_1}, \dots, a_{p_{k-1}}$, choose $p_k \in B_{\frac{1}{k}}(a)\setminus\{a\}$,
Then, similarly, there is $a_{p_k} \in B_{\frac{1}{k}}(a)$ such that $p_k > p_{k-1}$,

because no such p_k exists, then it is contradiction to a_{p_k} is a subsequential limit. Therefore, the arbitrary limit point a is contained in $\{a_n\}^*$,
So it is closed. ⌟

b). Since a), $\{a_n\}^*$ is closed and bounded, so compact. Thus

$$\sup\{a_n\}^* = \max\{a_n\}^* = a \text{ for some subsequential limit } a.$$

So, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that $n \geq N \Rightarrow a_n < a + \varepsilon$ (since $a = \max\{a_n\}^*$)
and there exists infinitely many terms in $\{a_n\}$ such that $a - \varepsilon < a_n$ (since $a_{n_k} \to a$)
Thus $\limsup_{n\to\infty} a_n = a$.

c). Let b_n be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given $b \in [0,1]$, there exists a subsequence in $\{b_n\}$ converges to b, because: for any $\varepsilon > 0$, there are infinitely many terms in b_n such that

$$|b_n - b| < \varepsilon,$$

So we can construct the subsequence. ☒

「2025, Fall.

17. Let E be a nonempty bounded subset of a Metric space (X,d). 10m

Define $\quad \operatorname{diam} E := \sup\{d(x,y) \mid x,y \in E\}$

a). Prove $\operatorname{diam} E = \operatorname{diam} \overline{E}$

b). If E is Compact, then $d(x',y') = \operatorname{diam} E$ for some $x', y' \in E$.

Proof.

a). Since $E \subseteq \overline{E}$, $\operatorname{diam} E \le \operatorname{diam} \overline{E}$ is clear.

To show the inverse inequality, using the Property of Supremum,

Given $\varepsilon > 0$, there exists $x, y \in \overline{E}$ such that

$$\operatorname{diam} \overline{E} - \frac{\varepsilon}{2} < d(x,y) \le \operatorname{diam} \overline{E}$$

And, since $\overline{E}$ is closure, there are $x', y' \in E$ such that

$$d(x',x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y,y') < \frac{\varepsilon}{4}.$$

Now, $\operatorname{diam} \overline{E} < d(x,y) + \frac{\varepsilon}{2} < d(x,x') + d(x',y') + d(y',y) + \frac{\varepsilon}{2} < d(x',y') + \varepsilon.$

Since $\varepsilon > 0$ was arbitrary, $\operatorname{diam} \overline{E} \le \operatorname{diam} E$, So Proved.」 7m.

b). Since E is Compact set, so $E \times E$ is also Compact set.

Since the metric function d is continuous, it Preserves compactness,

so $d[E \times E] \subseteq \mathbb{R}$ is Compact set. Therefore,

$$\operatorname{diam} E = \max d[E \times E] \in d[E \times E],$$

So there is $(x', y') \in E \times E$ such that $d(x',y') = \max d[E \times E]$.」 3m.

「2025. Fall.

8. Let $a>1$. Evaluate $\boxed{5m}$

$$\lim_{t\to0^+}\int_t^{at}\frac{\cos x}{x}\,dx$$

pruf. Consider the taylor expansion

$$\cos x=\sum_{n=0}^{\infty}(-1)^n\cdot\frac{x^{2n}}{(2n)!}$$

Therefore $\dfrac{\cos x}{x}=\dfrac{1}{x}-\dfrac{x}{2!}+\dfrac{x^3}{4!}-\cdots$

Now, $$\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\int_t^{at}-\frac{x}{2!}+\frac{x^3}{4!}-\cdots\,dx$$

$$=\ln a+\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}$$

(It is possible, because $\sum(-1)^n\cdot x^{2n-1}/(2n)!$ converges for all $x\in\mathbb{R}$)

Now, $\displaystyle\lim_{t\to0^+}\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx=0$, so

the given limit is $\ln a$.

Revised in Next.

「Observation for $\lim_{t\to0+}\int_t^{at}\frac{\cos x}{x}\,dx$.

As we know, the power series representation

$$\frac{1}{x}\cdot\cos x=\frac{1}{x}\cdot\sum_{n=0}^{\infty}(-1)^n\frac{x^{2n}}{(2n)!}=\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\frac{x^{2n-1}}{(2n)!}$$

is valid for all $x\neq0$. Therefore, for any $t>0$,

$$\int_t^{at}\frac{\cos x}{x}\,dx=\int_t^{at}\frac{1}{x}+\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

$$=\int_t^{at}\frac{1}{x}\,dx+\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\,dx$$

Meanwhile, the series converges uniformly in $[t,at]$ for small enough $t>0$, specifically, $t<\frac{1}{a}$, then for each $n\in\mathbb{N}$,

$$\left|(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}\right|\leq\left|\frac{1}{(2n)!}\right|\quad\text{and}\quad\sum_{n=1}^{\infty}\frac{1}{(2n)!}<\infty,$$

$\frac{1}{2n}x^{2n}$

therefore the Weierstrass M-test gives the condition.

Thus, we can exchange the integral to the sum,

$$\int_t^{at}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}\int_t^{at}(-1)^n\cdot\frac{x^{2n-1}}{(2n)!}=\sum_{n=1}^{\infty}(-1)^n\cdot\left(\frac{1}{2n\cdot(2n)!}\cdot\left(a^{2n}t^{2n}-t^{2n}\right)\right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\lim_{t\to0+}\sum_{n=1}^{\infty}(-1)^n\cdot\frac{1}{2n\cdot(2n)!}\cdot t^{2n}(a^{2n}-1)$$

$$=\sum_{n=1}^{\infty}\lim_{t\to0+}\quad //$$

$$=\sum_{n=1}^{\infty}0$$

$$=0$$

Consequently, the given integral is

$$\int_t^{at}\frac{1}{x}\,dx=\ln at-\ln t=\ln a.$$

Note: $\frac{\cos x}{x}=\frac{1}{x}+\frac{\cos x-1}{x}$, $\left|\int_t^{at}\frac{\cos x-1}{x}\,dx\right|\leq\int_t^{at}\frac{x}{2}\to0$ as $t\to0$.

「2025. Fall.

12. Define an equi. rel on $\mathbb{R}$ as $x \sim y \iff x-y \in \mathbb{Q}$.
 " as $x \approx y \iff x=y$ or $x, y \in \mathbb{Q}$.

a). Describe the quotient topology on $\mathbb{R}/\sim$.

b). " on $\mathbb{R}/\approx$.

c). In $\mathbb{R}/\approx$, determine if every point is closed

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since $\mathbb{Q}$ is closed under the operation $-$, so these relations are equi. rel.

a). First, $\mathbb{R}/\sim = \{a+\mathbb{Q} \mid a \in \mathbb{R}\}$, where $a+\mathbb{Q} = \{a+q \mid q \in \mathbb{Q}\}$. And, given a basis element $(a,b) \subseteq \mathbb{R}$, for the Projection $\pi : \mathbb{R} \to \mathbb{R}/\sim : a \mapsto a+\mathbb{Q}$.

$$\pi^{-1}[\pi[(a,b)]] = \bigcup_{r \in (a,b)} r+\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a+q, b+q)$$

is open and since π is quotient map, so $\pi[(a,b)]$ is open, so π is open.
Therefore, the Topology on $\mathbb{R}/\sim$ is generated by

$$\mathcal{B}_\sim = \{\pi[(a,b)] \mid a<b\} = \{(a,b)+\mathbb{Q} \mid a<b\}.$$

But, by the density of rationals, for any $a<b$, $(a,b)+\mathbb{Q} = \mathbb{R}/\mathbb{Q}$.
Therefore the Topology on $\mathbb{R}/\sim$ is indiscrete.

TSNU. Qual.

Prove that $\lim_{b\to\infty}\int_0^b \frac{\sin x}{x}\,dx$ exists.

Proof. First, let $a_n = n\pi$ $(n\in\mathbb{N})$. Then, a sequence $\left\{\int_0^{a_n}\frac{\sin x}{x}\,dx\right\}$ converges, because:

$\left|\int_{n\pi}^{(n+1)\pi}\frac{\sin x}{x}\,dx\right|$ is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi}\frac{\sin x}{x}\,dx$ is positive if n is even and is negative if n is odd.

That is, $\int_0^{n\pi}\frac{\sin x}{x}\,dx = \sum_{k=1}^{n}\int_{a_{k-1}}^{a_k}\frac{\sin x}{x}\,dx$ is alternating series, so converges.

「2026. Spring.

$\boxed{2nm}$

3. Let V be an n-dimensional vector space over $\mathbb{C}$.
Let $T: V \to V$ be a linear operator such that $T^3 = T$.
a). Find all possible characteristic polynomials and corresponding minimal polynomial for T.
b). Prove $V = \operatorname{Ker} T \oplus \operatorname{Im} T$.

Proof.

a). Since $T^3 - T = 0$, the minimal polynomial $m(t)$ for T divides $t^3 - t = 0$.
$t^3 - t = t(t-1)(t+1)$, there are possibilities:

$$m(t) = t \quad \text{or} \quad t-1 \quad \text{or} \quad t+1 \quad \text{or} \quad t(t-1) \quad \text{or} \quad t(t+1) \quad \text{or} \quad (t-1)(t+1) \quad \text{or} \quad t(t-1)(t+1).$$

Since $m(t)$ divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \quad \text{where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If $r_1 = 0$ and $r_2, r_3 \neq 0$, then the corresponding minimal polynomial is

$$(t-1)(t+1)$$

and other cases are similar.

b). ~~Given $\alpha \in \operatorname{Ker} T \cap \operatorname{Im} T$, $T\alpha = 0$ and $\alpha = T\beta$ for some $\beta \in V$.~~

Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and since $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$,
so $\alpha - T\alpha^2 \in \operatorname{Ker} T$ and trivially $T^2\alpha \in \operatorname{Im} T$. Moreover, if $T\alpha = 0$ and $\alpha \in \operatorname{Im} T$,
Then $\alpha = T\beta = T^3\beta$ for some $\beta \in V$ and $T\alpha = T^2\beta = 0$ so $T^3\beta = T\,T^2\beta = \alpha = 0$.
so $V = \operatorname{Ker} T \oplus \operatorname{Im} T$. ∎

[Modified.

Let T be a linear operator on the fin. dim space V over $\mathbb{C}$ satisfying $T^4 = T^2$.

a). Prove that $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Describe $T|_{\operatorname{Im} T^2}$.

c). Prove that T is diagonalizable $\Leftrightarrow$ $\ker T = \ker T^2$.

observe: $T^4 - T^2 = 0$, so the minimal polynomial divides $t^2(t-1)(t+1)$.

a). Given $\alpha \in V$, $\alpha = T^2\alpha + \alpha - T^2\alpha$ and $T^2\alpha \in \operatorname{Im} T^2$ clearly, and $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$, so $\alpha - T^2\alpha$, which proves $\alpha - T^2\alpha \in \ker T^2$, so $V = \ker T^2 + \operatorname{Im} T^2$. Moreover, given $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$, $T^2\alpha = 0$ and $\alpha = T^2\beta$ for some $\beta \in V$, so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

So $V = \ker T^2 \oplus \operatorname{Im} T^2$.

b). Given $\alpha \in \operatorname{Im} T^2$, $\alpha = T^2\beta$ for some $\beta \in V$. so, $T\alpha = T^3\beta$. Multiflying T, then $T^2\alpha = T^4\beta = T^2\beta$, so $T^2(\alpha - \beta) = 0$. There fore, $\alpha - \beta \in \ker T^2$, but since $\alpha \in \operatorname{Im} T^2$ and $\operatorname{Im} T^2 \cap \ker T^2 = 0$, $\beta = \alpha + \gamma$, where $\gamma \in \ker T^2$. Now, $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$, So, $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$. moreover, since $T^2 = T^4$, $T^2|_{\operatorname{Im} T^2} = I$.

c). Since T is diagonalizable $\Leftrightarrow$ minimal polynomial of T splits and separable. That is, the minimal polynomial is $t^a(t-1)^b(t+1)^c$, $a, b, c \in \{0, 1\}$.

「2026 Fall.

4. Let G be a group and $H_1, H_2 \leq G$ be subgroups of finite index.

a). Prove that the map

$$T : G \to S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and $\operatorname{Ker} T \subseteq H_1 \cap H_2$.

b). Prove $H_1 \cap H_2$ has finite index.

Proof

a). Given $g_1, g_2 \in G$, $T(g_1 g_2)(aH_1, bH_2) = (g_1 g_2 aH_1, g_1 g_2 bH_2)$

$$= T(g_1) \circ T(g_2)(aH_1, bH_2)$$

for each $a, b \in G$, so it preserves the operation.

And $g \in \operatorname{Ker} T \Leftrightarrow T_g = id \Leftrightarrow$ for all $a, b \in G$, $T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$

$\Leftrightarrow$ for all $a, b \in G$, $a^{-1} g a \in H_1$ and $b^{-1} g b \in H_2$

$\Rightarrow g \in H_1 \cap H_2$ (take $a = b = 1_G$)

b). By the first-isomorphism theorem,

$$G/\operatorname{Ker} T \cong \operatorname{Im} T \leq S_{G/H_1 \times G/H_2}$$

and since H_1, H_2 have finite index, so the codomain of T is finite set.

Now, $|G : H_1 \cap H_2| \leq |G : \operatorname{Ker} T| \leq \# S_{G/H_1 \times G/H_2}$

because $\operatorname{Ker} T \subseteq H_1 \cap H_2$, so proved.

Note: If $A \leq B \leq G$ and $|G : A|$ is finite, then

$$G/A \to G/B : gA \mapsto gB$$

is well-defined and onto, because: Since $A \subseteq B$,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

T2026. Fall.

8. For $f \in C([0,1])$, define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x<0 \\ f(x) & \text{if } 0 \le x \le 1 \\ f(1) & \text{if } x>1 \end{cases}$$

For each $n \ge 1$, define

$$\rho_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define $T_n f(x) := \int_{\mathbb{R}} \rho_n(x-y) \tilde{f}(y)\, dy, \quad x \in \mathbb{R}.$

a). Show that $T_n f \to f$ on $[0,1]$,

b). Determine $T_n f \to f$ uniformly on $[0,1]$ or not.

Proof. Given $x \in \mathbb{R}$ and $n \in \mathbb{N}$, $x-y \in [-\frac{1}{n}, \frac{1}{n}] \iff -\frac{1}{n}+x \le y \le \frac{1}{n}+x$. Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y)\, dy.$$

Since $\tilde{f}$ is continuous on $[-1,2]$, so it is continuously uniformly.

Given $\varepsilon > 0$, take $\delta > 0$ such that $\forall x, t \in [0,1]$, $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$.

Now, given $x \in [0,1]$, and $n > \frac{1}{\delta}$,

$$\begin{aligned} \left| T_n f(x) - f(x) \right| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y)\, dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot f(x)\, dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \cdot \left[\tilde{f}(y) - f(x) \right] dy \right| \\ &\le \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)|\, dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore, $\sup_{[0,1]} |T_n f - f| \to 0$ as $n \to \infty$, so converges uniformly.